

Mount Sinai, New York, NY, ⁴Department of Hematology and Medical Oncology, Icahn School of Medicine at Mount Sinai, New York, NY

Purpose/Objective(s): As the human papilloma virus (HPV) pandemic evolves and life expectancy increases, the number of older adults with oropharyngeal squamous cell carcinoma (OPSCC) continues to increase. However, there is a paucity of data regarding the fastest growing subset of this population: octogenarians (OGs). We sought to understand differences in treatment tolerability and clinical outcomes between septuagenarians (SGs) and OGs with OPSCC.

Materials/Methods: We identified SGs (age 70-79) and OGs (age 80-89) with OPSCC from a cohort of older adults with nonrecurrent, nonmetastatic head and neck squamous cell carcinoma (HNSCC) treated curatively from 2007-2020. We compared demographics, treatment characteristics and toxicities using Fischer's exact test. Time-to-event outcomes, overall survival (OS), locoregional control (LRC), and disease-specific survival (DSS), were evaluated using the Kaplan-Meier method.

Results: Of 293 patients (age 70-89) with HNSCC, 39% (n = 114) had OPSCC: 93 SGs (median age: 73; interquartile range [IQR]: 71-76), and 21 OGs (median age: 81, IQR: 80-84). The median follow-up for included patients was 2.4 years; 82% were male, 64% white, 48% $\geq$ 20 pack year smoking history, 37% ECOG 1. Patients had AJCC 8th edition Stage: I (27%); II (33%); III (18%); IV (22%) OPSCC. Treatment consisted of adjuvant radiation (RT) (19%), adjuvant chemoradiation (CRT) (8%), surgery alone (6%), induction/concurrent CRT (27%), concurrent CRT (28%), or RT (12%), with no significant differences in stage or treatment modalities noted between SGs and OGs. Of note, 69% of SGs and 76% of OGs were HPV+. Among 107 patients who received any RT, 24% experienced a treatment interruption (19% of SGs vs. 48% of OGs, $p=0.001$) and 2 patients (both HPV- SGs) died on treatment due to unrelated health conditions. Percutaneous endoscopic gastrostomy (PEG) tubes were placed prior to or during treatment in 43% of SGs and 62% of OGs, with OGs more likely to have a PEG placed during treatment ($p=0.025$). There was no difference in the prevalence of late ($\geq$ 6 months) CTCAE grade 2+ dysphagia (36%) or xerostomia (31%) between SGs and OGs. Estimated 3-year LRC, DSS, and OS were not significantly different between SGs (LRC:85%; DSS:87%; OS:76%) and OGs (LRC: 81%; DSS:94%; OS: 55%, p -values: 0.98, 0.42, 0.052, respectively). However, HPV+ disease significantly increased estimated 3-year OS for both SGs (HPV+: 84%; HPV-: 56%, $p=0.0006$) and OGs (HPV+: 68%; HPV-: 20%, $p=0.008$).

Conclusion: In our cohort, OGs had a higher proportion of HPV+ OPSCC, which was associated with improved OS. This finding may provide insight into the latency of the virus. While there were similar amounts of toxicities among SGs and OGs, OGs more frequently underwent PEG tube placement and experienced more treatment interruptions. Given high rates of HPV+ OPSCC in OGs, our findings suggest that de-escalation strategies should be further investigated to improve tolerability and maximize outcomes for this neglected population.

Author Disclosure: D.R. Dickstein: None. E.J. Lehrer: None. J.R. Bloom: None. K. Hsieh: None. B. Jones: None. J. Runnels: None. A. Powers: None. J. Barlow: None. S. Chen: None. E. Monroe: None. K. Sindhu: None. O. Factor: None. J.T. Liu: None. V. Gupta: None. S. Roof: None. D. Kirke: None. K. Misiukiewicz: None. M. Posner: None. E. Genden: None. R.L. Bakst: None.

3265

Cost-Effectiveness of Prophylactic Radiation in Patients with Asymptomatic Bone Metastases

A. Dornisch,^{1,2} and J.D. Murphy^{1,2}; ¹Department of Radiation Medicine and Applied Sciences, UC San Diego, La Jolla, CA, ²Center for Health Education and Research, University of California, San Diego, La Jolla, CA

Purpose/Objective(s): Radiation therapy represents a primary therapy for palliation of symptomatic bone metastases. Recently, a randomized trial found that radiation may prevent skeletal-related events (SREs) among patients with asymptomatic bone metastases. Skeletal-related events, such as pathologic fracture or spinal cord compression, cause considerable morbidity and decrements in quality of life. However, asymptomatic bone metastases are quite common and treating large numbers of these patients with radiation adds cost to the health care system. With this study we evaluated the cost-effectiveness of external beam radiation therapy for high-risk asymptomatic skeletal metastases.

Materials/Methods: We built a microsimulation model for cancer patients with high-risk asymptomatic bone metastasis who received either prophylactic external beam radiation therapy or standard of care. The model incorporated costs, quality of life (measured by health utility), as well as the risk of skeletal-related events, and risk of death. We measured cost-effectiveness with the incremental cost-effectiveness ratio (ICER), with ICERs under \$150,000 per quality-adjusted life-year (QALY) considered cost-effective. One way and probabilistic sensitivity analyses were used to test model uncertainty.

Results: Simulating the clinical course of 1,000,000 patients with asymptomatic bone metastases, we found that compared to standard of care, radiation therapy increased overall cost by \$22,431 and improved effectiveness by 0.37 QALYs, resulting in an ICER of \$60,400/QALY. The model was most sensitive to assumptions about survival and the monthly cost of cancer care. The base model assumed that prophylactic radiation reduced the risk of death by 50% (Hazard ratio [HR] 0.5). Prophylactic radiation remained cost-effective with HRs up to 0.98, though beyond an HR of 0.98 prophylactic radiation was no longer cost-effective. If the monthly cost of ongoing cancer care increased from our base estimate of \$1917 to \$5567, then prophylactic radiation was no longer cost effective. The model was not sensitive to assumptions about the cost of radiation, quality of life, or model time horizon. Probabilistic sensitivity analysis demonstrated that radiation therapy was cost-effective in >99% of iterations.

Conclusion: This study found that prophylactic radiation may represent a cost-effective treatment option by reducing the risk of skeletal-related events among patients with asymptomatic bone metastases. Longer follow-up and additional research defining efficacy and optimal target populations who benefit most from prophylactic radiation will help confirm these findings.

Author Disclosure: A. Dornisch: None. J.D. Murphy: None.

3266

Radiotherapy Quality Indicators in the Pilot Program of the NON in Poland in 2019-2021

A. Sztuder,¹ A.A. Drapiewski,² A. Maciejczyk,² and N. Wiśniewska²; ¹Department of Oncology, Medical University of Wrocław, Wrocław, Poland, ²Lower Silesian Oncology, Pulmonology and Hematology Center, Wrocław, Poland

Purpose/Objective(s): The pilot program of the National Oncology Network (NON) is the 1st Polish joint action concerning the construction of an oncological network, including the construction of a quality monitoring system. Data collection was started prospectively after obtaining consent from the Patient since February 2019 in 2 voivodeships: Lower Silesia and Holy Cross, and since October 2019 also in Podlaskie and Pomeranian. The aim of the pilot program was to assess the organization and quality of oncological care in selected voivodeships in 5 types of cancer – prostate, ovarian, colorectal, breast and lung cancer.

Materials/Methods: For the needs of the pilot program, an IT quality monitoring system was built, including: medical data warehouse, data reporting program and data visualization tool. The quality indicators (QI) were calculated in the Lower Silesia and those related to RT were further